# MSG No. 65.486 $Cmm'm'$ [ Type III, orthorhombic ]

Table 1: Wyckoff site: `2a`, site symmetry: `mm'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,2,3,4,5,6,7,8]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[9,10,11,12,13,14,15,16]` |

Table 2: Wyckoff site: `2b`, site symmetry: `mm'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | `[1,2,3,4,5,6,7,8]` |
| 2 | $\left[0, \frac{1}{2}, 0\right]$ | `[9,10,11,12,13,14,15,16]` |

Table 3: Wyckoff site: `2c`, site symmetry: `mm'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8]` |
| 2 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[9,10,11,12,13,14,15,16]` |

Table 4: Wyckoff site: `2d`, site symmetry: `mm'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[9,10,11,12,13,14,15,16]` |

Table 5: Wyckoff site: `4e`, site symmetry: `..2'/m'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[1,8,11,14]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[2,7,12,13]` |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | `[3,6,9,16]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[4,5,10,15]` |

Table 6: Wyckoff site: `4f`, site symmetry: `..2'/m'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[1,8,11,14]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[2,7,12,13]` |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[3,6,9,16]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[4,5,10,15]` |

Table 7: Wyckoff site: `4g`, site symmetry: `2m'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,2,7,8]` |
| 2 | $[-x,\ 0,\ 0]$ | `[3,4,5,6]` |
| 3 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[9,10,15,16]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$ | `[11,12,13,14]` |

Table 8: Wyckoff site: `4h`, site symmetry: `2m'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | `[1,2,7,8]` |
| 2 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | `[3,4,5,6]` |
| 3 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[9,10,15,16]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[11,12,13,14]` |

Table 9: Wyckoff site: `4i`, site symmetry: `m2'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ 0]$ | `[1,4,5,8]` |
| 2 | $[0,\ -y,\ 0]$ | `[2,3,6,7]` |
| 3 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$ | `[9,12,13,16]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$ | `[10,11,14,15]` |

Table 10: Wyckoff site: `4j`, site symmetry: `m2'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ y,\ \frac{1}{2}\right]$ | `[1,4,5,8]` |
| 2 | $\left[0,\ -y,\ \frac{1}{2}\right]$ | `[2,3,6,7]` |
| 3 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | `[9,12,13,16]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | `[10,11,14,15]` |

Table 11: Wyckoff site: `4k`, site symmetry: `mm'2'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4,6,7]` |
| 2 | $[0,\ 0,\ -z]$ | `[2,3,5,8]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | `[9,12,14,15]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | `[10,11,13,16]` |

Table 12: Wyckoff site: `4l`, site symmetry: `mm'2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,4,6,7]` |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 2 | $\left[0,\ \frac{1}{2},\ -z\right]$ | [2,3,5,8] |
| 3 | $\left[\frac{1}{2},\ 0,\ z\right]$ | [9,12,14,15] |
| 4 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | [10,11,13,16] |

Table 13: Wyckoff site: 8m, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$ | [1,14] |
| 2 | $\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$ | [2,13] |
| 3 | $\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$ | [3,16] |
| 4 | $\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$ | [4,15] |
| 5 | $\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$ | [5,10] |
| 6 | $\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$ | [6,9] |
| 7 | $\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$ | [7,12] |
| 8 | $\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$ | [8,11] |

Table 14: Wyckoff site: 8n, site symmetry: m..

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ z]$ | [1,4] |
| 2 | $[0,\ -y,\ -z]$ | [2,3] |
| 3 | $[0,\ y,\ -z]$ | [5,8] |
| 4 | $[0,\ -y,\ z]$ | [6,7] |
| 5 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$ | [9,12] |
| 6 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$ | [10,11] |
| 7 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$ | [13,16] |
| 8 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$ | [14,15] |

Table 15: Wyckoff site: 8o, site symmetry: .m'.

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ z]$ | [1,7] |
| 2 | $[x,\ 0,\ -z]$ | [2,8] |
| 3 | $[-x,\ 0,\ -z]$ | [3,5] |
| 4 | $[-x,\ 0,\ z]$ | [4,6] |
| 5 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ z\right]$ | [9,15] |
| 6 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | [10,16] |
| 7 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ -z\right]$ | [11,13] |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ z\right]$ | [12,14] |

Table 16: Wyckoff site: `8p`, site symmetry: `..m'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | `[1,8]` |
| 2 | $[x,\ -y,\ 0]$ | `[2,7]` |
| 3 | $[-x,\ -y,\ 0]$ | `[3,6]` |
| 4 | $[-x,\ y,\ 0]$ | `[4,5]` |
| 5 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$ | `[9,16]` |
| 6 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$ | `[10,15]` |
| 7 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ 0\right]$ | `[11,14]` |
| 8 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$ | `[12,13]` |

Table 17: Wyckoff site: `8q`, site symmetry: `..m'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ y,\ \frac{1}{2}\right]$ | `[1,8]` |
| 2 | $\left[x,\ -y,\ \frac{1}{2}\right]$ | `[2,7]` |
| 3 | $\left[-x,\ -y,\ \frac{1}{2}\right]$ | `[3,6]` |
| 4 | $\left[-x,\ y,\ \frac{1}{2}\right]$ | `[4,5]` |
| 5 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | `[9,16]` |
| 6 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | `[10,15]` |
| 7 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | `[11,14]` |
| 8 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | `[12,13]` |

Table 18: Wyckoff site: `16r`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[x,\ -y,\ -z]$ | `[2]` |
| 3 | $[-x,\ -y,\ -z]$ | `[3]` |
| 4 | $[-x,\ y,\ z]$ | `[4]` |
| 5 | $[-x,\ y,\ -z]$ | `[5]` |
| 6 | $[-x,\ -y,\ z]$ | `[6]` |
| 7 | $[x,\ -y,\ z]$ | `[7]` |
| 8 | $[x,\ y,\ -z]$ | `[8]` |
| 9 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$ | `[9]` |
| 10 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$ | `[10]` |
| 11 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$ | `[11]` |
| 12 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$ | `[12]` |
| 13 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$ | `[13]` |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$ | `[14]` |
| 15 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$ | `[15]` |
| 16 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$ | `[16]` |